INTERIM PROJECT REPORT

Predictive Maintenance

Business Problem, Data Registration, EDA, Data Preparation, Model Building and Experiment Tracking

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Dataset: engine_data.csv

Report type: Interim Project Report

This report summarizes the project workflow, evidence, preliminary findings and business implications. Raw code and full execution evidence are provided in the accompanying executed notebook and HTML export.

1. Executive Summary

This project evaluates whether six engine operating measurements can distinguish the recorded *Engine Condition* labels. The supplied dataset contains 19,535 records and 7 columns. Initial validation reported zero missing cells and zero duplicate rows.

Four classifiers were compared using stratified three-fold cross-validation on the training data. AdaBoost achieved the highest mean cross-validation weighted F1 score (0.6462) and was selected. On the held-out test set, it achieved 65.42% accuracy, 0.6385 weighted F1 and 0.6846 ROC-AUC.

> **Interpretation caution:** The supplied material does not define whether labels 0 and 1 represent failure, normal operation or another state. These results demonstrate classification performance on this dataset; they do not establish advance failure prediction or production readiness.

2. Business Problem Understanding

Business context. Unplanned equipment downtime may increase maintenance costs and disrupt reliability. Telemetry analytics may help identify patterns that warrant investigation when the target label and operating context are defined.

Problem statement. Determine whether engine RPM, lubricating-oil pressure, fuel pressure, coolant pressure, lubricating-oil temperature and coolant temperature contain information useful for classifying the recorded Engine Condition label.

Objectives. Validate the data; explore target balance, feature distributions and relationships; prepare a reproducible training/test split; compare tuned classification models; and document findings and limitations.

Scope. This interim report covers data registration through experimentation tracking. Deployment and quantified financial benefits are outside the current scope.

3. Data Registration

Methodology. The dataset is stored in the repository at `data/engine_data.csv`. The validation script checks expected columns, non-empty input, missing values, binary target values, row/column counts, duplicates and target distribution.

Table 1. Dataset validation summary

Data check	Result
Rows	19,535
Columns	7
Predictors	6
Missing cells	0
Duplicate rows	0
Target labels	0 and 1

Table 2. Dataset fields and roles

Column	Role / interpretation
Engine rpm	Numeric engine measurement; units and acceptable ranges require domain confirmation.
Lub oil pressure	Numeric engine measurement; units and acceptable ranges require domain confirmation.
Fuel pressure	Numeric engine measurement; units and acceptable ranges require domain confirmation.
Coolant pressure	Numeric engine measurement; units and acceptable ranges require domain confirmation.
lub oil temp	Numeric engine measurement; units and acceptable ranges require domain confirmation.
Coolant temp	Numeric engine measurement; units and acceptable ranges require domain confirmation.
Engine Condition	Target label; operational meaning of 0 and 1 needs confirmation.

The initial checks support proceeding to EDA but do not prove that sensor values are physically valid. Sensor units, acceptable ranges and the label-generation process should be confirmed with a domain expert.

4. Exploratory Data Analysis (EDA)

Methodology. EDA used target counts and proportions, descriptive statistics, class-wise feature distributions and feature-to-target correlation. The supplied report values are reproduced below; patterns are descriptive and do not imply causation.

4.1 Univariate analysis: target distribution

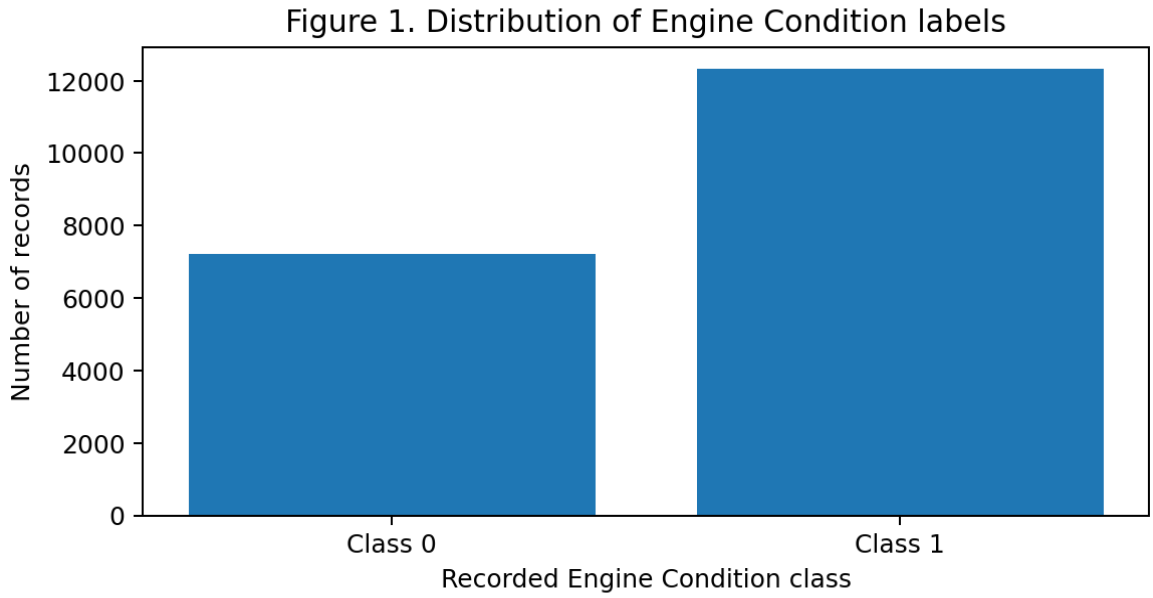


Figure 1. Distribution of Engine Condition labels. Source: counts reported in the supplied project report.

Table 3. Target class counts and proportions

Class	Count	Share
0	7,218	36.95%
1	12,317	63.05%

Class 1 accounts for 63.05% of records and Class 0 for 36.95%. The distribution is uneven, so evaluation should include per-class measures and comparison with a baseline rather than accuracy alone.

4.2 Univariate analysis: feature summary statistics

Table 4. Summary statistics for numeric features

Feature	Mean	Std. dev.	Min	Median	Max
Engine rpm	791.24	267.61	61.00	746.00	2239.00
Lub oil pressure	3.30	1.02	0.00	3.16	7.27
Fuel pressure	6.66	2.76	0.00	6.20	21.14
Coolant pressure	2.34	1.04	0.00	2.17	7.48
lub oil temp	77.64	3.11	71.32	76.82	89.58
Coolant temp	78.43	6.21	61.67	78.35	195.53

Summary statistics show scale, spread and possible extreme values. For example, the reported maximum coolant temperature (195.53) is much higher than its median (78.35). This warrants checking against source documentation and sensor specifications; it should not be removed automatically without domain validation.

4.3 Bivariate analysis: feature distributions by class

The supplied report includes class-wise histograms for engine RPM, lubricating-oil pressure, fuel pressure and coolant temperature. The distributions overlap, suggesting that individual measurements may not fully separate the recorded classes. The notebook should be used to inspect these plots at full resolution; their interpretation requires validation with quantitative class-level metrics.

> Business implication: no single feature should be treated as a stand-alone maintenance rule based on these plots. A multivariable model may capture combined patterns, but the meaning of the target labels and measurement ranges must be confirmed first.

4.4 Multivariate analysis: feature-to-target correlation

The report provides Pearson correlations between each numeric feature and the target. This is a feature-wise correlation summary, not a complete pairwise correlation matrix.

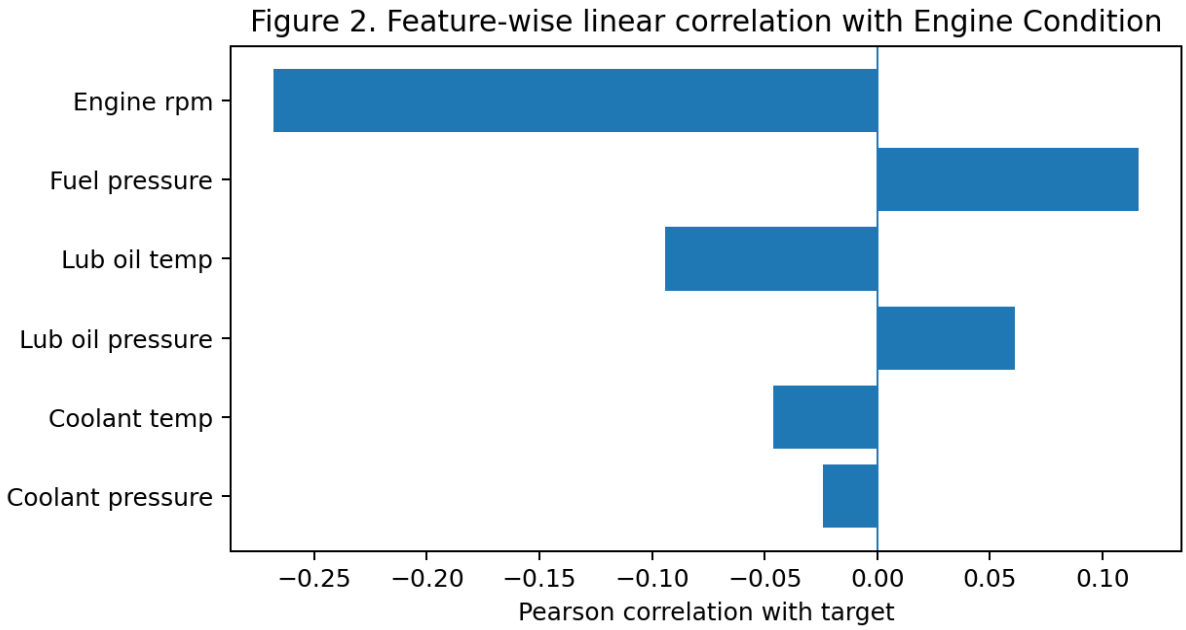


Figure 2. Feature-wise linear correlation with Engine Condition. Values reproduced from the supplied report.

Table 5. Feature-wise correlation with target

Feature	Correlation with target
Engine rpm	-0.268
Fuel pressure	0.116
lub oil temp	-0.094
Lub oil pressure	0.061
Coolant temp	-0.046
Coolant pressure	-0.024

Engine RPM has the largest absolute linear correlation with the target (–0.268), which is modest. Its direction cannot be translated into failure risk until label meanings are confirmed. Correlation can also miss non-linear relationships and interactions.

5. Data Preparation

Methodology. The dataset was split reproducibly using stratified 80/20 sampling and a fixed random seed. The training set contains 15,628 records and the held-out test set contains 3,907 records. Stratification approximately preserves the class proportions.

Table 6. Train/test split

Split	Rows	Purpose
Training set	15,628	Fit candidate models and perform cross-validation tuning.
Test set	3,907	Final evaluation after model selection.

The train and test files are saved as workflow artifacts and are also present in the repository under `artifacts/splits/`. No scaling was required for the tree-based models evaluated. If records repeat by asset or time, an asset-aware or time-aware split may better reflect real-world generalization; this should be checked with the data owner.

6. Model Building and Experiment Tracking

Methodology. A majority-class Dummy Classifier provides a baseline. Decision Tree, Random Forest, AdaBoost and Gradient Boosting were tuned using GridSearchCV with stratified three-fold cross-validation. Mean weighted F1 was the primary model-selection metric. The held-out test set was not used to select the winning model.

Table 7. Cross-validation model comparison and selected parameters

Candidate model	Best CV weighted F1	Best parameters reported
AdaBoost	0.6462	learning_rate=1.0; n_estimators=20
Gradient Boosting	0.6375	learning_rate=0.1; max_depth=2; n_estimators=40
Random Forest	0.6350	n_estimators=40; max_depth=None; min_samples_leaf=1
Decision Tree	0.6078	criterion=gini; max_depth=8; min_samples_leaf=1

AdaBoost was selected because it achieved the highest mean cross-validation weighted F1 (0.6462). The experiment-tracking output records parameter combinations, cross-validation scores, training scores, ranks and fit-time information.

6.1 Held-out test evaluation

Table 8. Test-set evaluation results reported for candidate models and baseline

Model	Accuracy	Precision (weighted)	Recall (weighted)	F1 (weighted)	F1 (macro)	ROC-AUC
AdaBoost	0.6542	0.6384	0.6542	0.6385	0.6002	0.6846
Gradient Boosting	0.6660	0.6499	0.6660	0.6336	0.5861	0.6981
Random Forest	0.6424	0.6254	0.6424	0.6266	0.5873	0.6645
Decision Tree	0.6063	0.6449	0.6063	0.6131	0.6009	0.6537
Dummy baseline	0.6304	0.3974	0.6304	0.4875	0.3867	N/A

Gradient Boosting has the highest reported test accuracy (66.60%) and ROC-AUC (0.6981), but AdaBoost remains selected because the predefined selection rule uses training-only cross-validation weighted F1. The test results are reported as final evaluation evidence and should not be used to retroactively change model selection.

6.2 Selected model: class-level results

Table 9. AdaBoost classification report

Class / average	Precision	Recall	F1-score	Support
Class 0	0.5453	0.3878	0.4533	1,444
Class 1	0.6931	0.8104	0.7471	2,463
Macro average	0.6192	0.5991	0.6002	3,907
Weighted average	0.6384	0.6542	0.6385	3,907

AdaBoost recalls Class 1 more often than Class 0. If one class represents a critical maintenance state, this difference may have operational consequences. Label semantics must be confirmed before assigning a cost to either error type. Review the notebook's confusion matrix and ROC curve with the mentor.

7. Preliminary Business Insights and Recommendations

Methodology. Recommendations below link the descriptive checks and evaluation outputs to follow-up actions. They are provisional because the supplied materials do not define the target-label semantics, sensor units, asset identifiers or prediction horizon.

Table 10. Business insights and recommended actions

Observation	Business implication	Recommended action
No missing cells or duplicate rows were reported.	Basic data-quality checks passed, but physical validity is not established.	Confirm sensor units, acceptable ranges and sensor-health rules with the data owner.
Class 0 is 36.95%; Class 1 is 63.05%.	A majority-class baseline can appear accurate without useful per-class discrimination.	Use per-class metrics and compare model performance with the baseline.
AdaBoost has the highest CV weighted F1 (0.6462).	It is the leading candidate under the predefined selection metric, not yet production-ready.	Validate on new, appropriately separated data and review errors by class.
AdaBoost Class 0 recall is 0.3878.	A substantial share of Class 0 records are not identified as Class 0.	Confirm label meaning and determine the business cost of false positives and false negatives.
Label definitions and time/asset context are undocumented in the reviewed material.	Advance-warning and operational claims remain uncertain.	Request label rules, timestamps, asset IDs and the intended prediction horizon.

8. Limitations and Questions for Mentor / Data Owner

- What do labels 0 and 1 mean operationally, and how were they assigned?
- What are the sensor units and acceptable operating ranges?
- Are records repeated by engine/asset or time, and can timestamps or asset IDs be provided?
- Is the objective to classify current state or predict failure a defined time before it occurs?
- Which error is more costly: missed maintenance risk or unnecessary inspection?
- What business-defined performance threshold and external validation set should be used?

> The dataset appears to be a static labelled dataset in the reviewed columns. Model results are preliminary and should not be used alone to trigger maintenance decisions.

9. Conclusion and Next Steps

The project establishes a repeatable workflow from data validation through EDA, train/test preparation, model tuning and experiment tracking. AdaBoost was selected using training-only cross-validation, with mean weighted F1 of 0.6462; held-out test accuracy was 65.42% and ROC-AUC was 0.6846.

Next steps are to confirm target-label definitions and sensor ranges, review confusion-matrix and class-level errors with the mentor, and evaluate the selected model on appropriately separated data before discussing deployment or financial benefits.

Appendix A. Reproducibility and Supporting Evidence

Raw code and full execution outputs are provided in the accompanying executed Jupyter notebook and its HTML export. The notebook and repository artifacts should be used to verify the model, metadata, experiment-tracking log, evaluation table and train/test split files.

Representative repository paths: `data/engine_data.csv`; `src/validate_data.py`; `src/prepare_data.py`; `src/train_model.py`; `artifacts/splits/train.csv`; `artifacts/splits/test.csv`; and `artifacts/model/`.

Validation and workflow evidence. The GitHub Actions workflow was observed to complete both jobs successfully: data validation/split creation and model tuning/evaluation. The run produced workflow artifacts for splits and model results. This is pipeline execution evidence; it is not proof of production readiness.

Appendix B. Sources and Data Provenance

Dataset: *engine_data.csv*, the file supplied with the project materials. The original public dataset URL or publication details were not included in the materials reviewed for this revision. Add the original source and access date here if known; do not cite an unverified source.

Analysis and metrics in this report are reproduced from the supplied interim report and project notebook/artifacts. External methodological sources should be added if external text, definitions, datasets or figures are introduced.

End of Interim Project Report